



PREDICTING STUDENT DROPOUT RISK IN HIGHER EDUCATION: A MACHINE LEARNING APPROACH

Predicción de riesgo de deserción estudiantil en educación superior: Un enfoque en aprendizaje automático

Andres Peralta-Fragozo 
Universidad Nacional de Colombia, Valledupar, Colombia. 
aperaltaf@unal.edu.co

Juan-Pablo Hoyos-Sanchez 
Universidad Nacional de Colombia, Valledupar, Colombia. 
jhoyoss@unal.edu.co

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ABSTRACT

Recently, student dropout has been a major challenge for higher education institutions, since high dropout rates generate negative consequences, both academic and social, as well as economic losses for all involved. For these reasons, this study proposes a machine learning-based approach for the early prediction of dropout risk, based on academic, socioeconomic, and demographic variables. The methodology started with an exploratory data analysis, followed by a class balancing of the dataset using the SMOTE technique. Subsequently, supervised classification algorithms such as Support Vector Machines (SVM), XGBoost, Random Forest, and Balanced Random Forest Classifier were trained to build the prediction models. GridSearchCV was used to search for the best hyperparameters. The models were evaluated using metrics robust to class imbalance, such as recall, ROC-AUC curve, geometric mean (G-Mean), precision, and F1-Score. The best-performing models were Random Forest and its Balanced Random Forest Classifier variant, without hyperparameter optimization, with AUC values of 0.94 and recall values greater than 0.8 in the positive class (at-risk students). In addition, model explanation with SHAP values was used to identify and analyze the most influential risk factors (academic and socioeconomic characteristics).

Keywords: Balanced Random Forest Classifier; classification; higher education; machine learning; Random Forest; SMOTE; student dropout; SVM; XGBoost.

RESUMEN

Recientemente la deserción estudiantil ha venido representando grandes desafíos para las instituciones de educación superior, ya que las altas tasas de deserción generan consecuencias negativas, tanto académicas como sociales, así como pérdidas económicas para todos los involucrados. Por tales motivos, este estudio propone un enfoque basado en aprendizaje automático para la predicción temprana del riesgo de deserción, a partir de variables académicas, socioeconómicas y demográficas. Se propone una metodología que inicia con un análisis exploratorio de los datos seguido de un balanceo de las clases del conjunto de datos mediante la técnica de SMOTE. Posteriormente se entrenan los algoritmos de clasificación supervisada como, Support Vector Machines (SVM), XGBoost, Random Forest y Balanced Random Forest Classifier para la construcción de los modelos de predicción. Para la búsqueda de los mejores hiperparámetros se utilizó la técnica de búsqueda en malla (GridSearchCV). Los

modelos fueron evaluados considerando métricas robustas al desbalance de clases, como el recall, la curva ROC-AUC, la media geométrica (G-Mean), precisión y F1-Score. Los modelos que mejor desempeño obtuvieron fueron Random Forest y su variante Balanced Random Forest Classifier, sin optimización de hiperparámetros con valores de AUC de 0.94 y recall superiores al 0.8 en la clase positiva (estudiantes en riesgo). Además, se utilizó la explicación de modelos con valores SHAP para identificar y analizar los factores de riesgo más influyentes (características académicas y socioeconómicas).

Palabras clave: aprendizaje automático; Balanced Random Forest Classifier; clasificación; deserción estudiantil; educación superior; Random Forest; SMOTE; SVM; XGBoost.

PREDIÇÃO DE RISCO DE EVASÃO ESTUDANTIL NO ENSINO SUPERIOR: UMA ABORDAGEM BASEADA EM APRENDIZADO DE MÁQUINA

RESUMO

Recentemente, a evasão estudiantil tem representado grandes desafios para as instituições de ensino superior, pois as altas taxas de abandono geram consequências negativas, tanto acadêmicas quanto sociais, além de perdas econômicas para todos os envolvidos. Por esse motivo, este estudo propõe uma abordagem baseada em aprendizado de máquina para a predição precoce do risco de evasão, a partir de variáveis acadêmicas, socioeconômicas e demográficas. Propõe-se uma metodologia que inicia com uma análise exploratória dos dados, seguida do balanceamento das classes do conjunto de dados por meio da técnica SMOTE. Posteriormente, são treinados algoritmos de classificação supervisionada, como Support Vector Machines (SVM), XGBoost, Random Forest e Balanced Random Forest Classifier, para a construção dos modelos de predição. Para a busca dos melhores hiperparâmetros, utilizou-se a técnica de busca em grade (GridSearchCV). Os modelos foram avaliados considerando métricas robustas ao desbalanceamento de classes, como recall, curva ROC-AUC, média geométrica (G-Mean), precisão e F1-Score. Os modelos que apresentaram melhor desempenho foram Random Forest e sua variante Balanced Random Forest Classifier, sem otimização de hiperparâmetros, com valores de AUC de 0,94 e recall superiores a 0,8 na classe positiva (estudantes em risco). Além disso, utilizou-se a explicação dos modelos com valores SHAP para identificar e analisar os fatores de risco mais influentes (características acadêmicas e socioeconômicas).

Palavras-chave: aprendizado de máquina; Balanced Random Forest Classifier; classificação; ensino superior; evasão estudiantil; Random Forest; SMOTE; SVM; XGBoost.

1. INTRODUCTION

Student academic performance is a fundamental pillar in higher education, especially in critical areas such as mathematics, which provide the foundation for engineering and science programs. The ability to anticipate performance in advanced courses by analyzing achievement in prerequisite subjects and personal factors allows educational institutions to design strategic interventions that improve pass rates and reduce student dropout [1]. In this context, machine learning (ML) algorithms gain relevance, as they enable the early identification of at-risk students and, consequently, the design of support or tutoring programs that foster meaningful learning [2].

Indeed, with the adoption of machine learning in the educational sector, the way large volumes of data are processed, analyzed, and visualized has been transformed, since these algorithms facilitate the identification of patterns and variables that are key to understanding academic performance [3]. In the state of the art, decision tree-based algorithms stand out for being robust and easily interpretable, as they expose the reasoning process that leads to their predictions [4]. This interpretability, that is, the ability for predictions to be interpreted by humans, has helped universities build early warning systems

and make real-time adjustments to their teaching and learning processes, enabling continuous and more effective support for their students [3, 5]. Recently, this emphasis on explainability has become a point of agreement within the community. It is no longer enough to have only an accurate model, as interpretability is essential to understanding why the model makes its decisions. This has driven the use of Explainable AI (XAI), with techniques such as SHAP and SAGE, to analyze complex models such as XGBoost or Random Forest, extracting truly useful conclusions for institutional decision-making [6-7].

One fact that cannot be overlooked is that the rise of digital platforms has made an unprecedented amount of data available to educational researchers, thus opening new avenues for improving various pedagogical approaches [2-3]. An example of new approaches can be seen in studies that use Game Learning Analytics to analyze interaction data, revealing that behavior-related variables have greater predictive value [8]. Similarly, other studies examine students' behavior using recreational applications, finding that students who spent more time planning achieved higher academic performance [9]. These types of findings suggest an important lesson: if more comprehensive predictive models are desired, it is beneficial to incorporate qualitative and behavioral data [10]. In fact, a common conclusion in the literature is that the most robust models are those that integrate the student's academic history, personal context, socioeconomic level, and even their place of origin [2, 3, 11].

Given the need to process this multiplicity of variables efficiently, Random Forest (RF) has become a particularly popular algorithm for these types of complex scenarios [12]. This algorithm is based on constructing multiple decision trees from subsets of data (bootstrap) and making decisions by averaging the predictions or by applying "democracy" depending on whether the context is regression or classification, which gives it stability and reliability, especially when working with a large number of attributes [12].

Although RF and other models have remarkable qualities, this does not necessarily mean that they are a universal solution, since the effectiveness of the classifier varies depending on the data set and the specific problem being addressed [13-14]. For example, class imbalance is a recurring challenge that has motivated researchers to explore technical combinations of oversampling, among which SMOTE stands out, achieving effective balancing and remarkable accuracy [14]. This highlights the importance of carefully selecting ML algorithms. While simpler models, such as Naive Bayes or rule-based models, can be useful, they often struggle to handle the inherent complexity of social and imbalanced data. In educational contexts, dropout students frequently represent a minority proportion, which tends to bias model performance [3, 15].

It's important to highlight that other lines of research have focused on unsupervised learning techniques, such as clustering methods (BIRCH, DBSCAN), to identify risk profiles based on the digital footprint that students leave on virtual learning platforms [16-17].

In this context, the present study proposes a comprehensive approach for the early prediction of dropout risk through the integration of ML and explainable analysis. Our framework incorporates four essential components. First, the construction of a hybrid target variable based on data-driven criteria and administrative records of confirmed dropout, ensuring statistical robustness and ethical validity. Second, the handling of class imbalance using SMOTE and specialized RF variants such as the Balanced Random Forest Classifier. Third, a rigorous comparison of several algorithms evaluated with metrics robust to imbalance. And fourth, the application of XAI techniques (SHAP) to identify the most influential risk factors, thereby making the decision-making process transparent. This approach not only seeks to maximize predictive performance but also produces interpretable and accessible models that allow institutions to implement evidence-based early warning systems.

2. METHODOLOGY

2.1. Overview of the approach

To implement the proposed methodological framework, a dataset provided by the Universidad Nacional de Colombia, La Paz campus, was used for the period 2019-2024. It contained 1,445 records and 52 features. With the aim of reducing dimensionality, improving model performance, and minimizing the risk of overfitting, a cleaning process that included the elimination of irrelevant, non-informative variables and unique identifiers was carried out. Thus, the dataset was reduced to 1,294 records and 23 features (Table 1), consolidating relevant administrative and sociodemographic information and providing a comprehensive representation of the student academic profile. The proposed workflow is illustrated in Figure 1, which covers the stages of preprocessing and exploratory analysis, imbalance handling, training, and interpretability analysis. The following subsections describe each component in detail.

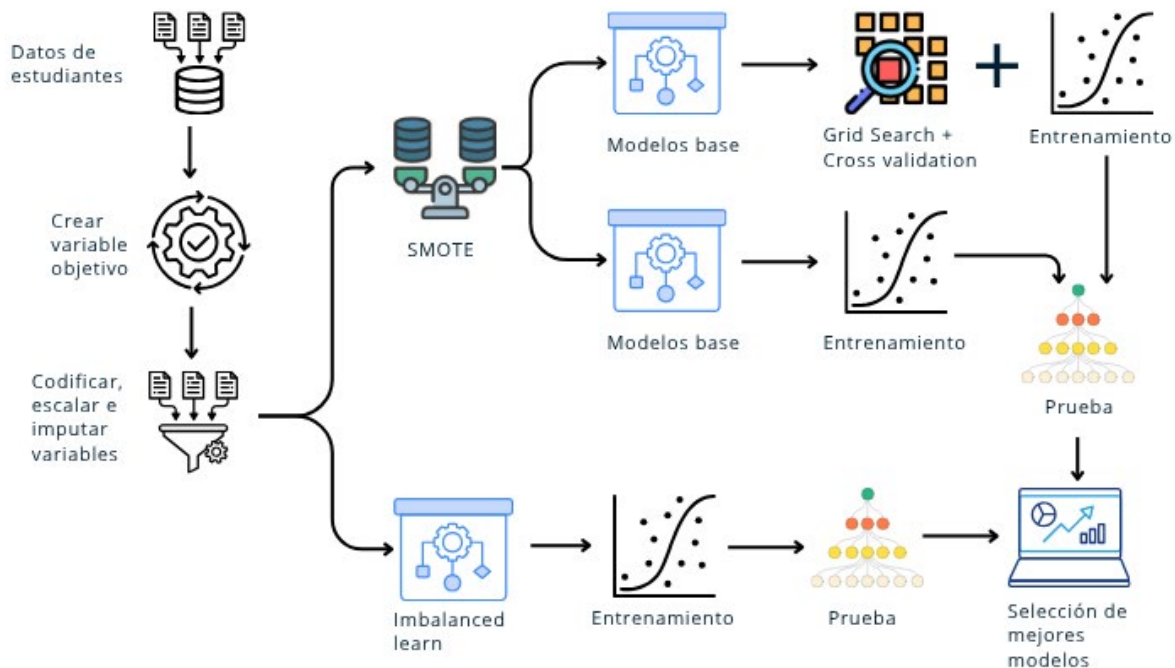


Figure 1. Proposed workflow in the methodology.

Table 1. Feature selection for predictive modeling.

Features	Description
Continuous numeric	
AVANCE_CARRERA	Percentage of progress in the curriculum
EDAD	Student's age
PAPA	Cumulative weighted arithmetic mean (Grade Point Average)
PROME_ACADE	Student's overall academic average
PBM_CALCULADO	Baseline Enrollment Score
PUNTAJE_ADMISION	Score obtained on the admission exam
AVANCE_ESPERADO	Expected progress based on enrollment numbers
ATRASO_PORCENTUAL	Percentage difference between expected and actual progress
Ordinal Integer Variables	
NUMERO_MATRICULAS	Number of academic periods enrolled
ESTRATO	Student socioeconomic status
Categorical	
PLAN	Student academic program
SUBACCESO	Subtype of university access
APERTURA	Start of academic period
MUNICIPIO_RESIDENCIA	Student's municipality of residence
MUNICIPIO_NACIMIENTO	Student's municipality of birth
DISCAPACIDAD	Type of disability reported
CARACTER_COLEGIO	Type of school of origin
MUNICIPIO_RESIDENCIA_FAM	Municipality of family residence
NODO_INICIO	Start node of the academic program
Binary	
GENERO	Student's biological gender
NACIONALIDAD	Student nationality
VICTIMAS_DEL_CONFLICTO	Victim status in armed conflict
Target Variable	
RIESGO_DESERCION	Risk indicator (0: no risk, 1: at risk)

2.2. Construction of the target variable

The target variable, RIESGO_DESERCION, was constructed using a hybrid approach that combines risk criteria derived from data analysis with validation based on administrative records. First, rules were defined through a contextual analysis of academic and socioeconomic indicators. The variable ATRASO_PORCENTUAL was created, and after analyzing its skewed distribution (see [Figure 2](#)), the 80th percentile was established as a threshold for identifying significant academic lag. Based on this, the “risk” label (1) was assigned by applying the following three specific criteria: (1) Students with more than 4 enrollments were labeled at risk if their percentage lag exceeded the 80th percentile and their PAPA was below 3.6; (2) Students with 4 or fewer enrollments were labeled at risk if their PAPA was below 3.6, prioritizing early detection before lag became evident; (3) New entrants (without PAPA) were evaluated according to their socioeconomic vulnerability, being labeled “at risk” if they belonged to stratum zero or one with a PBM below 3.

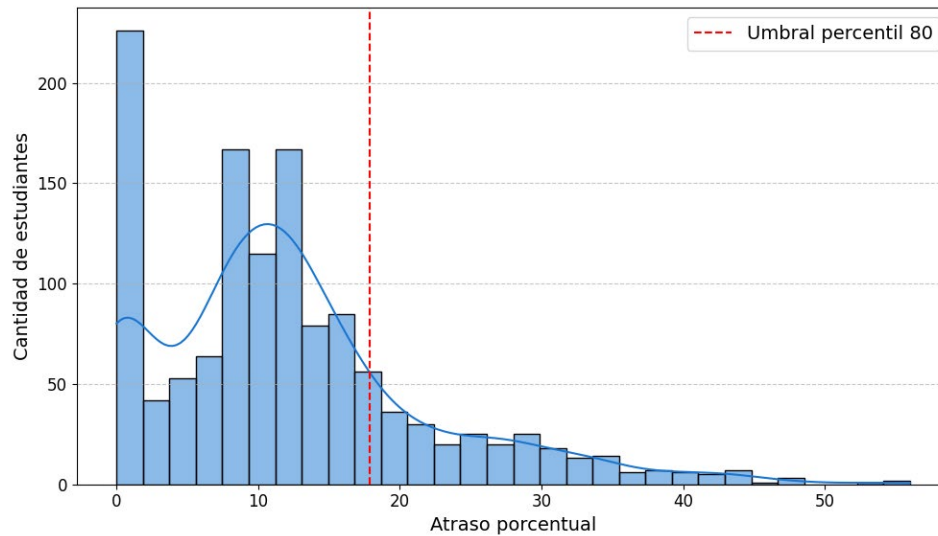


Figure 2. Distribution of percentage delay and the 80th percentile defined as delay threshold.

Afterward, this set of labels was enriched and validated using an administrative record containing 44 confirmed dropout cases. This critical step ensured that the 25 students from this group who had not been captured by the initial rules were also assigned to the at-risk class. Consequently, the final target variable consisted of 188 positive cases, representing 14.6% of the sample. Therefore, this hybrid design is crucial, as it provides a methodologically robust and ethically sound definition of risk by combining predictive patterns with already confirmed dropout data.

2.3. Exploratory analysis and data processing

An exploratory analysis was conducted to guide data preprocessing, including the imputation of missing values, distribution analysis, and interpretation of outliers. The handling of missing data was adapted to the proportion and nature of each variable. Features such as MUNICIPIO_RESIDENCIA, with extreme missingness (~70%), were removed and replaced by MUNICIPIO_RESIDENCIA_FAM. For PUNTAJE_ADMISION (27% missing), the mean by academic program was imputed to preserve its contextual variability. The variables PAPA and PROME_ACADE, with 7.2% missing values and high correlation, were imputed using MICE, after which PROME_ACADE was eliminated to mitigate multicollinearity.

The descriptive analysis of categorical variables revealed a strong imbalance in the disability variable, where most students (~99%) reported no condition. Despite its low frequency, this feature was retained due to its relevance from an inclusion perspective, as it enables the identification of critical cases. Likewise, it is noteworthy that approximately 10% of students identified themselves as victims of the armed conflict, a condition of structural vulnerability considered relevant for the model.

Likewise, the Spearman correlation analysis shown in Figure 3 confirmed a strong negative relationship between ATRASO_PORCENTUAL and PAPA (-0.59), indicating that performance decreases as academic lag increases. Moderate positive correlations were also observed between PUNTAJE_ADMISION and PAPA (0.22), as well as between PUNTAJE_ADMISION and PBM_CALCULADO (0.30), linking academic performance and socioeconomic context from the starting point.

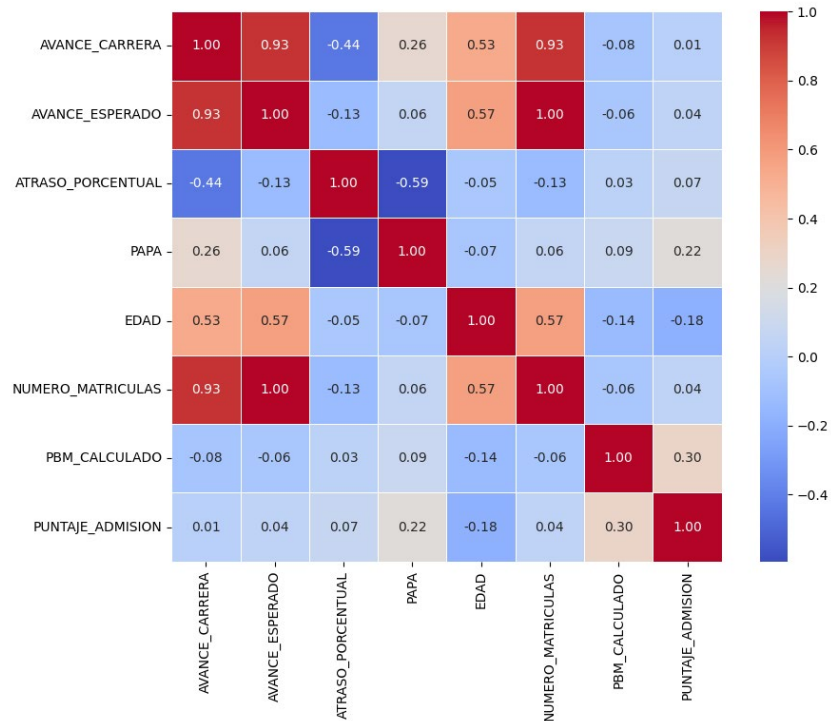


Figure 3. Spearman correlation matrix of the numerical features of the dataset.

On the other hand, the analysis of outliers using boxplots shown in Figure 4 determined that the extreme observations were informative, as they represent exceptional cases rather than data errors. The distribution of ATRASO_PORCENTUAL, which exhibited strong right skewness, contained outliers corresponding to cases of severe academic lag, thereby justifying the use of the 80th percentile as a threshold. Similarly, the outliers observed in PBM_CALCULADO (low scores) and EDAD (older ages) reflected conditions of socioeconomic vulnerability and non-traditional trajectories; therefore, they were retained.

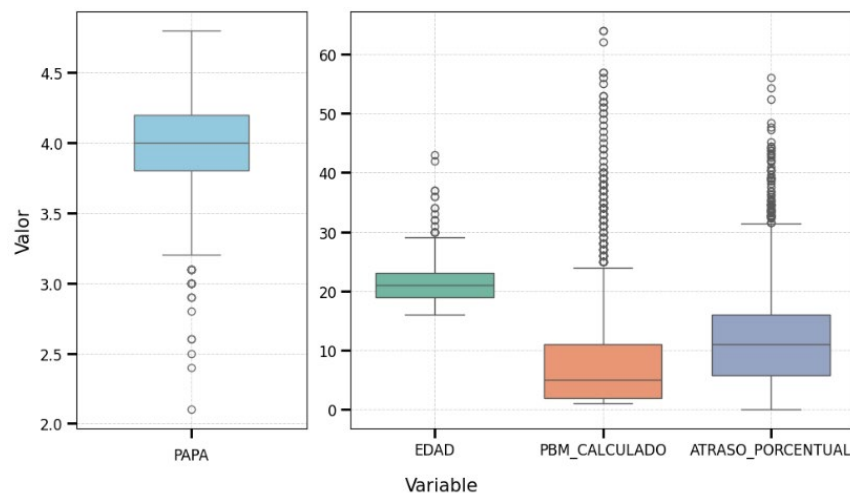


Figure 4. Distribution of numerical variables and detection of outliers using boxplots.

2.4. Selection and transformation of variables

To ensure model stability, the multicollinearity among numerical variables was evaluated using the Variance Inflation Factor (VIF). This analysis revealed severe collinearity ($VIF > 10$) in variables such as AVANCE_CARRERA and ATRASO_PORCENTUAL, and perfect collinearity in AVANCE_ESPERADO and NUMERO_MATRICULAS. Consequently, the latter variables were removed due to their redundancy, which stabilized the VIF levels of the remaining features to acceptable levels.

Subsequently, variable encoding and scaling were performed. Categorical features were transformed according to their cardinality. Target Encoding was applied to high-cardinality variables (more than six categories) to capture their relationship with the target variable without generating excessive dimensionality, while One-Hot Encoding or binary encoding was used for the remaining features. On the other hand, different scaling techniques were applied to the numerical variables, based on their distributions shown in Figure 5. RobustScaler was used for variables with outliers, such as ATRASO_PORCENTUAL, and PowerTransformer was used to correct bias in others. This preparation was crucial for optimizing the performance of the classification algorithms.

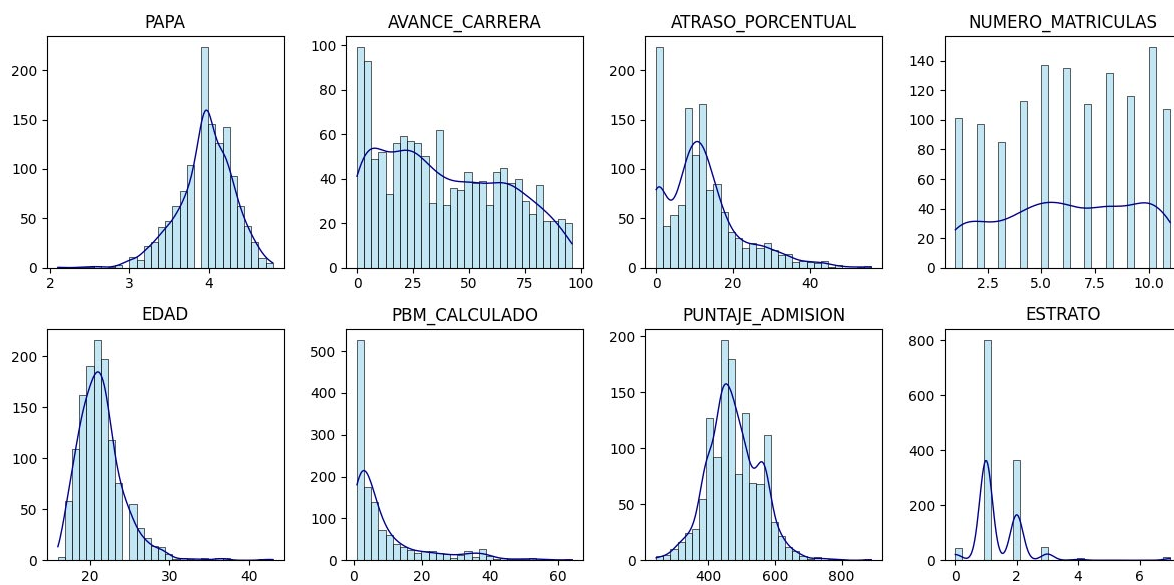


Figure 5. Distribution of numerical variables.

2.5. Handling class imbalance and evaluation metrics

The target variable, RIESGO_DESERCION, exhibited moderate imbalance, with only 14.6% of observations belonging to the risk class (class 1). To mitigate potential classifier bias toward the majority class, the SMOTE oversampling technique was applied to the training set. The Balanced Random Forest Classifier (BRFC) was excluded from this preprocessing step, as it internally manages imbalance through an undersampling mechanism. Given this distribution, the performance evaluation focused on metrics

robust to imbalance. Instead of relying solely on accuracy, priority was given to the area under the ROC curve (AUC), the geometric mean (G-Mean), and the recall of the positive class, as these allow for a more balanced assessment of the models' overall ability to correctly identify students at risk [18-19].

3. RESULTS

3.1. Performance and interpretability of the baseline models

Table 2 presents the results of the baseline models without hyperparameter optimization, evaluating their performance under class imbalance using metrics that measure the ability to effectively identify the minority class. This ensures a balance between sensitivity and precision in predicting dropout risk. To guarantee the reproducibility of this study, the source code, analysis notebooks, and processed data are available in the following GitHub repository: <https://github.com/acperaltaf/university-dropout-prediction.git>.

Table 2. Comparison of performance metrics - baseline models.

Model	Class	Precision	Recall	F1	Accuracy	G-Mean	AUC
Random Forest	0	0.967	0.993	0.980	0.966	0.893	0.948
	1	0.957	0.803	0.873			
XGBoost	0	0.960	1.000	0.980	0.960	0.874	0.950
	1	0.980	0.770	0.860			
SVM	0	0.950	0.940	0.940	0.900	0.807	0.930
	1	0.650	0.700	0.670			
Balanced Random Forest Classifier	0	0.970	0.970	0.970	0.950	0.892	0.958
	1	0.820	0.820	0.820			

The Random Forest and Balanced Random Forest classifiers demonstrated the most robust performance in identifying students at risk. Both achieved AUC values above 0.94 and a recall for the positive class exceeding 80%, effectively balancing sensitivity and precision. The high discriminative capacity of these models was confirmed by their respective ROC curves and confusion matrices (see Figure 7), which show correct classification of the vast majority of cases. In contrast, although XGBoost achieved competitive results (AUC of 0.95), its recall was slightly lower (0.77). The SVM model, for its part, showed greater difficulty handling class imbalance, registering the lowest recall (0.70).

Beyond predictive performance, the interpretability analysis revealed the key factors associated with student dropout. The feature importance results from the Random Forest model (Figure 6a) highlighted that the academic performance variables PAPA, ATRASO_PORCENTUAL, and AVANCE_CARRERA are the most influential predictors. The SHAP analysis of the BRFC model (Figure 6b) corroborated this finding and clarified the nature of the relationship: low values of PAPA and AVANCE_CARRERA drastically increase the probability of dropout, whereas high values reduce it. These results confirm that academic progress is the main factor in predicting risk, making these models viable and transparent tools for early detection.

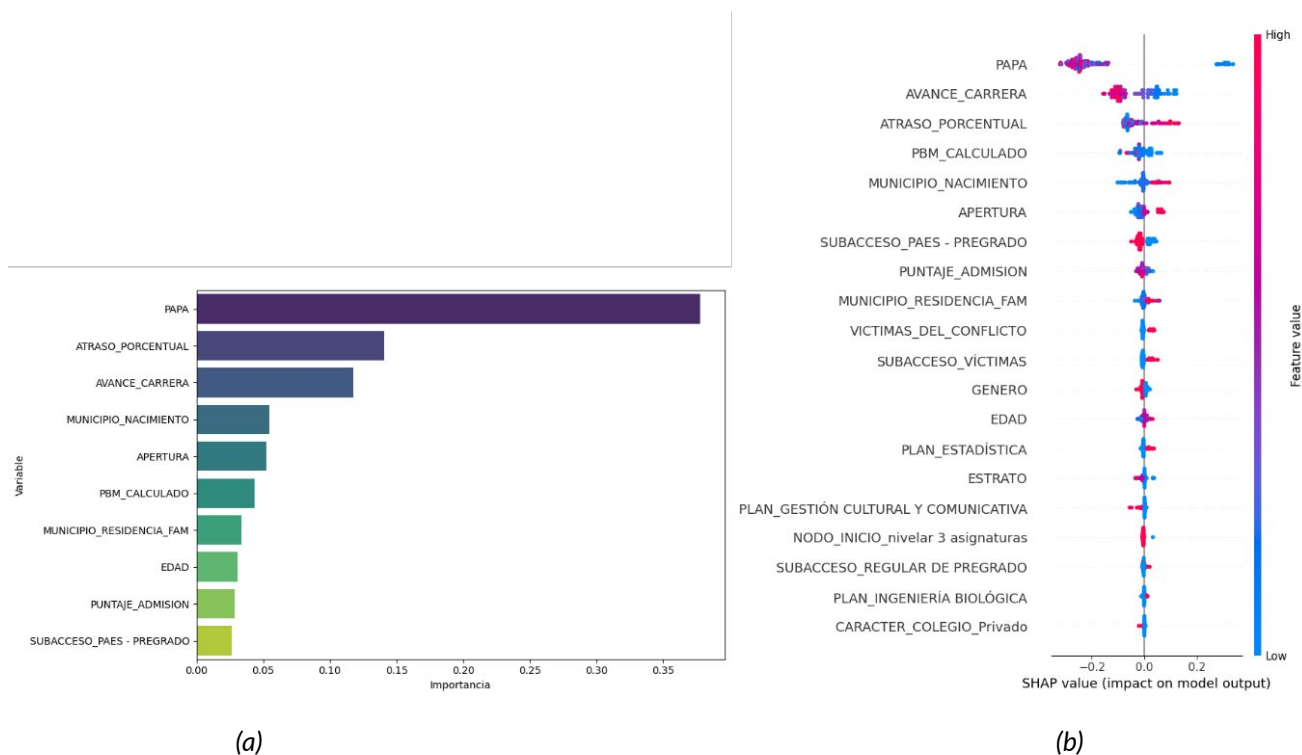


Figure 6. (a) Feature Importance of the Baseline Random Forest. (b) SHAP Values of the Baseline Balanced Random Forest Classifier.

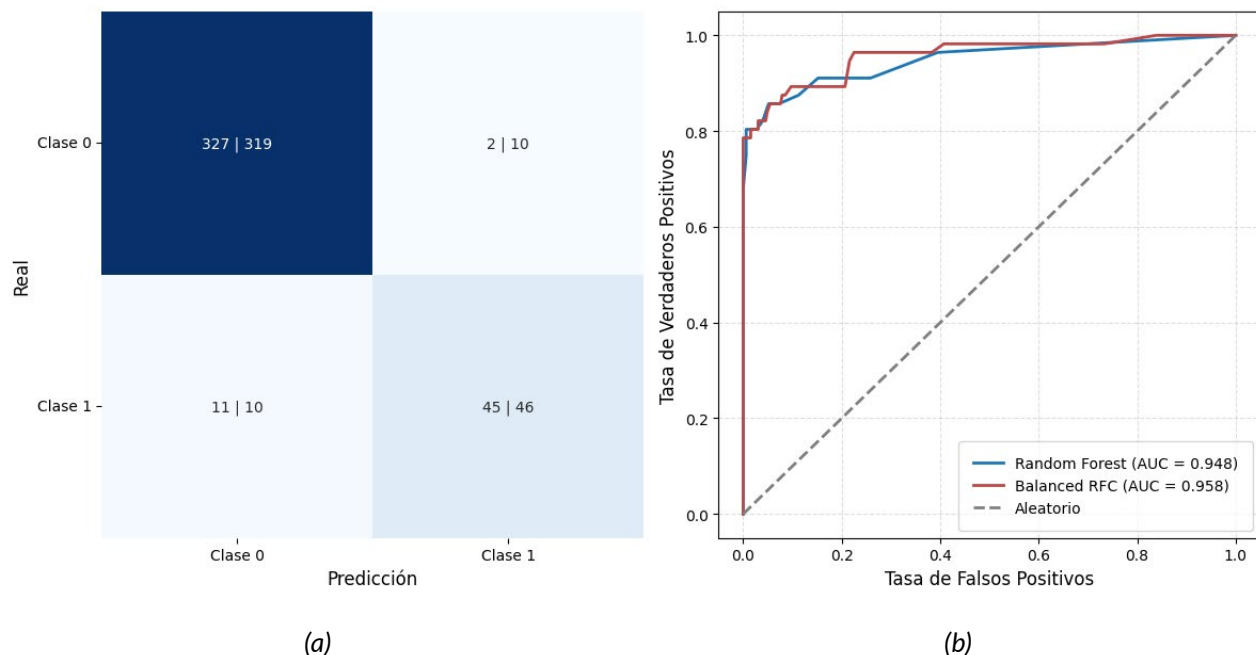


Figure 7. Comparison of confusion matrices and ROC curves of the Random Forest and Balanced Random Forest Classifier models.

3.2. Hyperparameter optimization with grid search and cross-validation

Hyperparameter optimization was performed using GridSearchCV, evaluating combinations of predefined parameters for each algorithm. Table 3 presents the tuned parameters and the range of values explored during the grid search, with the aim of identifying the configuration that maximizes performance while considering the imbalanced nature of the dataset.

Table 3. Hyperparameters evaluated with grid search.

Model	Parameters	Values
Random Forest	n_estimators	50, 100, 200
	max_depth	None, 5, 10, 20
	min_samples_split	2, 5, 10
	min_samples_leaf	1, 2, 4
	max_features	'sqrt', 'log2'
XGBoost	n_estimators	20, 50, 100, 150
	max_depth	3, 5, 7, 9
	learning_rate	0.01, 0.1, 0.2, 0.4
	subsample	0.6, 0.8, 1.0
	colsample_bytree	0.6, 0.8, 1.0
SVM	C	0.1, 1, 10, 100
	kernel	'poly', 'rbf', 'linear'
	gamma	'scale', 'auto'
	degree	2, 3, 4, 5

However, the results presented in Table 4 revealed that the performance improvements were mostly marginal compared with the baseline models. For example, the optimized Random Forest experienced a slight reduction in recall (from 0.80 to 0.77) in exchange for higher precision, while the Balanced Random Forest, despite a minor improvement in recall, suffered a notable decrease in its AUC (from 0.958 to 0.892). Meanwhile, the performance of SVM remained stable.

Table 4. Performance metrics of the models with hyperparameter search via Grid Search.

Models + GridSearch	Class	Precision	Recall	F1	Accuracy	G-Mean	AUC
Random Forest	0	0.960	1.000	0.980	0.960	0.874	0.954
	1	0.980	0.770	0.860			
XGBoost	0	0.970	0.980	0.980	0.960	0.889	0.943
	1	0.900	0.800	0.850			
SVM	0	0.940	0.950	0.950	0.910	0.791	0.911
	1	0.690	0.660	0.670			
Balanced Random Forest Classifier	0	0.970	0.950	0.960	0.930	0.890	0.892
	1	0.720	0.840	0.780			

Based on these results, it is reasonable to conclude that the default configurations of the ensemble models already offered a performance close to optimal for the data. Thus, although the optimization did not yield substantial improvements, it proved to be a useful step to verify the robustness of the selected models, confirming that their strong performance is consistent and not dependent on a specific hyperparameter calibration.

4. DISCUSSION

The methodology and results obtained in this study, focused on early dropout prediction through the integration of ML and XAI, align with and reinforce previous findings in the fields of Educational Data Mining and Learning Analytics aimed at identifying behavioral patterns and providing explanations to support targeted interventions [3, 7].

The results show a high potential for adapting machine learning techniques to enable early detection of dropout risk. One of the most notable findings was the development of ensemble models based on Random Forest, which demonstrated strong predictive performance, achieving AUC values of 0.94 and 0.96 in its balanced version using standard configurations, that is, without requiring a dedicated optimization stage.

However, we contrasted the impact of hyperparameter optimization. While this study found that tuning offered only marginal improvements in precision and recall, the literature has documented that metaheuristic optimization can significantly outperform base models in dropout prediction tasks [6, 14]. This disparity suggests that the effectiveness of optimization may be dataset-specific [17] or that ensemble algorithms, when trained on well-processed data, may already operate near optimal configurations.

Beyond predictive accuracy alone, the SHAP value analysis offered insights into the most influential variables driving the models' decisions. The results showed that the true predictive strength lies in variables capturing academic progress, such as PAPA, academic advancement, and accumulated delay, while also highlighting the valuable contribution of features related to psychological support, institutional characteristics, and socioeconomic factors, which are highly relevant in this context [3]. These findings further validate the practical potential of the proposed methodology for predicting student dropout.

Ultimately, the results reaffirm the viability of predictive models as practical tools, enabling institutions to implement proactive monitoring systems based on academic indicators. Since these models can "see" risk patterns long before a student decides to drop out, they open a crucial window for designing effective and timely interventions [2-16].

At the same time, the observation that socioeconomic factors also exert an influence, albeit to a lesser extent, reinforces the need for comprehensive support in intervention strategies, such as peer tutoring and professional psychological counseling.

5. CONCLUSIONS AND FUTURE WORK

This article presents a structured methodology that includes robust data preprocessing, statistical analysis, strategies for handling imbalanced data through SMOTE, and model optimization using a grid search for hyperparameters. The Spearman correlation analysis found a negative relationship between the variable *ATRASO_PORCENTUAL* and *PAPA* (-0.59), indicating that academic performance decreases as academic delay increases.

In the evaluation of the models, traditional metrics such as precision, recall, F1-score, and accuracy were used. However, since the distribution of the target variable is imbalanced, with only 14.6% of students classified as at risk, additional metrics suitable for these scenarios were included, namely the Geometric Mean (G-Mean) and AUC. Among the results obtained, the model with the best performance was the Balanced Random Forest, achieving an AUC of 0.958 and a G-Mean of 0.89. The interpretability analysis was carried out using SHAP, identifying the academic performance variables *PAPA*, *ATRASO_*

PORCENTUAL, and AVANCE_CARRERA as the most influential predictors. It was also found that socioeconomic factors have a smaller influence. Therefore, a comprehensive intervention is desirable, including options such as peer-tutoring, mentorship, and professional psychological support.

Although this study is limited to the dataset under analysis, the methodology provides a replicable approach for other educational institutions, which could integrate these models into automated pipelines for deploying them as services for large-scale use.

Future research could explore the incorporation of new strategies for handling imbalanced data, such as SMOGN, as well as hyperparameter optimization through genetic or Bayesian search methods, which would enhance the exploration and exploitation of new machine learning models. Additionally, working with larger datasets, including psychological, motivational, or family-related variables, as well as more complex architectures from areas such as deep learning, could allow the identification of more complex patterns than those captured by the current models.

AUTHORS' CONTRIBUTION

Andres Peralta-Fragozo: Data curation, Conceptualization, Investigation, Methodology, Writing–Review and Editing.

Juan-Pablo Hoyos-Sanchez: Conceptualization, Supervision, Methodology, Writing–Review.

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